Business Intelligence Use Case Proposal

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Project Title

Analysis of Digital Payment Adoption in Tunisia: A Business Intelligence Case Study Using Flouci

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1. Industry and Context

The financial services industry is undergoing a rapid digital transformation worldwide, driven by the rise of fintech solutions such as digital wallets, mobile payments, and cashless transaction platforms. Digital payment systems improve transaction efficiency, reduce dependency on cash, enhance financial inclusion, and enable better monitoring of economic activity.

In Tunisia, despite high mobile phone penetration and increasing internet usage, the adoption of digital payment solutions remains relatively low compared to global benchmarks. Cash is still the dominant means of payment, especially in small businesses and informal sectors. In response, banks, fintech startups, and regulators have introduced several digital payment solutions such as Flouci, D17, e-Dinar, and mobile banking wallets.

Flouci is a Tunisian digital wallet that enables peer-to-peer (P2P) transfers, merchant payments, bill payments, and mobile top-ups. It represents a suitable case study to analyze digital payment adoption patterns, user behavior, and transaction performance in Tunisia.

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2. Organization Description

The focus organization in this project is **Flouci**, a Tunisian fintech platform providing digital wallet services. Flouci targets individuals, merchants, and service providers by offering secure, fast, and accessible cashless payment solutions through mobile applications.

Key stakeholders include: - Flouci management and strategy teams - Merchants accepting digital payments - End users (individual customers) - Financial institutions and regulators

These stakeholders rely on data-driven insights to improve adoption, optimize user experience, expand merchant networks, and support national financial inclusion objectives.

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3. Business Problem

Despite the increasing availability of digital payment and wallet solutions in Tunisia, user adoption and long-term retention remain inconsistent. Many customers register for digital payment services but

become inactive or close their accounts after a short period. This phenomenon reflects challenges such as lack of trust, limited perceived value, income constraints, and uneven service penetration across demographic segments.

Using customer-level behavioral and demographic data (age, income, education, tenure, product category, and attrition status), fintech providers such as Flouci face difficulties in answering key questions: - Which customer segments are more likely to abandon digital payment services? - How do demographic and socio-economic factors influence adoption and retention? - Which product tiers (wallet categories) perform best in terms of customer longevity?

The absence of consolidated analytical dashboards makes it difficult to proactively reduce customer attrition and increase sustainable adoption. A Business Intelligence solution is therefore required to analyze customer behavior, identify high-risk segments, and support data-driven strategies for improving digital payment adoption in Tunisia.

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4. Business Objectives

The main objectives of this BI project are to: - Analyze digital payment adoption trends in Tunisia - Understand user behavior and transaction patterns - Identify factors influencing adoption and transaction success - Compare Tunisian digital payment usage with global benchmarks - Provide actionable insights to improve digital wallet adoption and performance

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5. Scope and Data Assumptions

This project uses an anonymized customer behavior dataset containing more than 10,000 records, sourced from a global digital payment and credit transaction dataset. The dataset includes key demographic, product, and behavioral variables such as customer age, gender, income category, education level, product category, customer tenure (Months_on_book), and attrition status (Attrition_Flag).

Due to limited access to large-scale public digital wallet transaction data in Tunisia, the dataset is adapted to the Tunisian context using demographic and financial inclusion indicators published by the World Bank. Flouci is used as a representative digital wallet to contextualize customer behavior, product tiers, and retention patterns within the Tunisian fintech ecosystem. All assumptions and transformations will be clearly documented to ensure transparency and academic validity.

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6. Key Analytical Questions

1. What is the overall customer attrition rate for digital payment users?
2. How does attrition vary by age group and gender?
3. Which income categories show the highest and lowest retention rates?
4. How does customer tenure (Months_on_book) influence attrition?
5. Which wallet or product categories (Basic, Standard, Premium, Platinum) retain users the longest?
6. How does education level impact digital payment adoption and retention?
7. Are customers with more dependents more likely to stop using digital payment services?

8. What demographic profiles represent high-risk (high attrition) customers?
9. Which customer segments should be prioritized for retention strategies?
10. How does customer behavior in Tunisia compare to global digital payment patterns?

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7. Key Performance Indicators (KPIs)

1. Customer Attrition Rate (percentage of customers with Attrition_Flag = 1)
2. Active Customer Rate
3. Average Customer Tenure (Months_on_book)
4. Attrition Rate by Age Group
5. Attrition Rate by Income Category
6. Attrition Rate by Education Level
7. Customer Distribution by Product / Wallet Category
8. Retention Rate by Product Tier

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8. Expected Business Value

This BI solution will provide Flouci and similar fintech stakeholders with a comprehensive view of digital payment adoption in Tunisia. The insights generated will support: - Improved marketing and user acquisition strategies - Better targeting of under-served regions and segments - Optimization of transaction processes - Data-driven policy and regulatory discussions

Ultimately, the project contributes to understanding how digital payments can accelerate financial inclusion and economic digitization in Tunisia.

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9. Conclusion

This project proposes a complete Business Intelligence solution to analyze digital payment adoption in Tunisia using Flouci as a case study. By integrating large-scale transactional data, dimensional modeling, KPIs, and analytical insights, the project aligns fully with the course requirements and demonstrates the practical value of BI in the fintech sector.